

Business Analytics Project

Repurposing the Familiar: ML-Guided Search for Antibiotics in FDA-Approved Drugs

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**“the most fruitful basis of the discovery of
a new drug is to start with an old drug”**

Nobel laureate Sir James Black



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Abbreviations

AMR	Antimicrobial Resistance
CRE	Carbapenem-Resistant Enterobacteriaceae
Da	Dalton (unit of molecular mass)
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
E. coli	<i>Escherichia coli</i>
ECFP4	Extended Connectivity Fingerprints (radius 2, diameter 4)
EHEC	Enterohemorrhagic <i>Escherichia coli</i>
ETEC	Enterotoxigenic <i>Escherichia coli</i>
FDA	Food and Drug Administration
GDP	Gross Domestic Product
HTS	High-Throughput Screening
k-NN	k-Nearest Neighbors
logP	Logarithm of the Partition Coefficient (octanol–water)
ML	Machine Learning
NPs	Natural Products
PCA	Principal Component Analysis
RDKit	Open-source cheminformatics toolkit
SAR	Structure–Activity Relationship
SHAP	SHapley Additive exPlanations
SMEs	Small and Medium-sized Enterprises
SMILES	Simplified Molecular Input Line Entry System
STEC	Shiga Toxin–Producing <i>Escherichia coli</i>
TPSA	Topological Polar Surface Area
UPEC	Uropathogenic <i>Escherichia coli</i>
UMAP	Uniform Manifold Approximation and Projection

Executive Summary

The global rise of antimicrobial resistance has created an urgent demand for novel antibiotic agents. While traditional discovery pipelines remain resource-intensive and slow, data-driven approaches offer a promising avenue for accelerating the identification of therapeutic candidates. This project explores a machine learning-based framework for antibiotic discovery through drug repurposing, focusing on FDA-approved non-antibiotic compounds. The aim was to leverage structural and physicochemical similarities with known antibiotics to uncover hidden antibiotic potential in existing drugs.

The methodology combined unsupervised and supervised learning strategies. Initially, molecules were represented using global physicochemical descriptors such as molecular weight, logP, and functional groups. These were projected into a two-dimensional chemical space using Uniform Manifold Approximation and Projection (UMAP), allowing for visualization and neighborhood analysis. A k-Nearest Neighbors (k-NN) classifier was then trained on both antibiotics and non-antibiotics within this reduced space, enabling the prediction of antibiotic-likeness probabilities for each non-antibiotic compound.

To further capture substructural information, unavailable in global physicochemical descriptors, Extended Connectivity Fingerprints (ECFP4) were computed. Using Tanimoto distances between fingerprint vectors, Density-Based Spatial Clustering (DBSCAN) was applied to identify structurally coherent clusters. A hybrid scoring system was devised by combining the k-NN antibiotic-likeness probability with a binary indicator of cluster membership, thereby balancing topological proximity with fingerprint-based similarity.

The k-NN approach yielded four high-confidence antibiotic-like candidates: metaxalone, gabapentin, enacarbil, primaquine, and lorcaserin. Of these, metaxalone was also prioritized by the hybrid scoring approach, reinforcing its status as a promising repurposing candidate. This overlap highlights how integrating global and local chemical features improves the robustness of candidate selection.

In summary, this work demonstrates the feasibility of using interpretable, structure-based machine learning techniques to discover novel antibiotic-like candidates. The proposed framework balances chemical insight with algorithmic simplicity, offering a reproducible and scalable tool for early-stage antibiotic discovery and drug repurposing.

Introduction

Background

“...But I would like to sound one note of warning. Penicillin is to all intents and purposes non-poisonous so there is no need to worry about giving a penicillin overdose and poisoning the patient. There may be a danger, though, in underdosage. It is not difficult to make microbes resistant to penicillin in the laboratory by exposing them to concentrations not sufficient to kill them...”.¹ This warning, raised by Alexander Fleming during his 1945 Nobel Prize acceptance speech for the discovery of penicillin (alongside Howard Florey and Ernst Chain), proved to be remarkably prescient in light of today’s silent pandemic of antibiotic resistance.

When people are prescribed an antibiotic, they expect a simple, two-step process: take the drug and get better. Such is the power and blessing that antibiotics have bestowed upon modern medicine that any other outcome feels almost unthinkable. Behind this smooth path from sickness to recovery lies a dire and silent reality. In the ongoing evolutionary battle for survival, bacteria are adapting faster than we are responding.

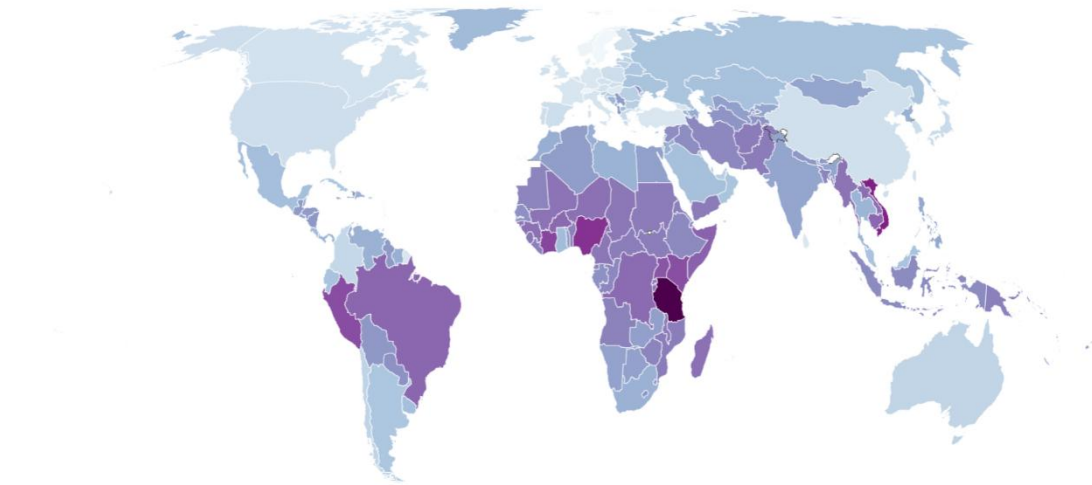
According to a report published in *The Lancet*,² an estimated 4.95 million deaths were associated with bacterial antimicrobial resistance (AMR) in 2019, including 1.27 million deaths directly attributable to it (approximately 3479 lives lost every single day). AMR occurs when microorganisms such as bacteria, viruses, fungi, and parasites evolve to resist the effects of the drugs designed to kill them, making infections increasingly difficult or even impossible to treat.³

Crucially, AMR is not driven solely by the misuse or overuse of antibiotics. It stems from a complex interplay of clinical, societal, and systemic factors, including public health policies, sanitation infrastructure, education, and access to diagnostics and medicines.^{4,5} As a result, low- and middle-income countries (LMICs) bear a disproportionate share of the burden. Paradoxically, the countries with the lowest availability of antibiotics, typically the poorest, are often those where antibiotic-resistant bacteria are most prevalent.

In these regions, the lack of access to effective antibiotics still poses a greater immediate threat than resistance itself.⁵ However, the large number of untreated, undertreated, or unsuccessfully treated infections creates ideal conditions for resistance to emerge and spread. This is evident in both the expression patterns of AMR genes⁵ (**Figure 1**) and in the elevated mortality rates attributed to AMR in these settings² (**Figure 2**). In other words, while AMR is a global issue, its most devastating impacts are concentrated in regions least equipped to respond.

AMR gene prevalence*

119.9 755.1



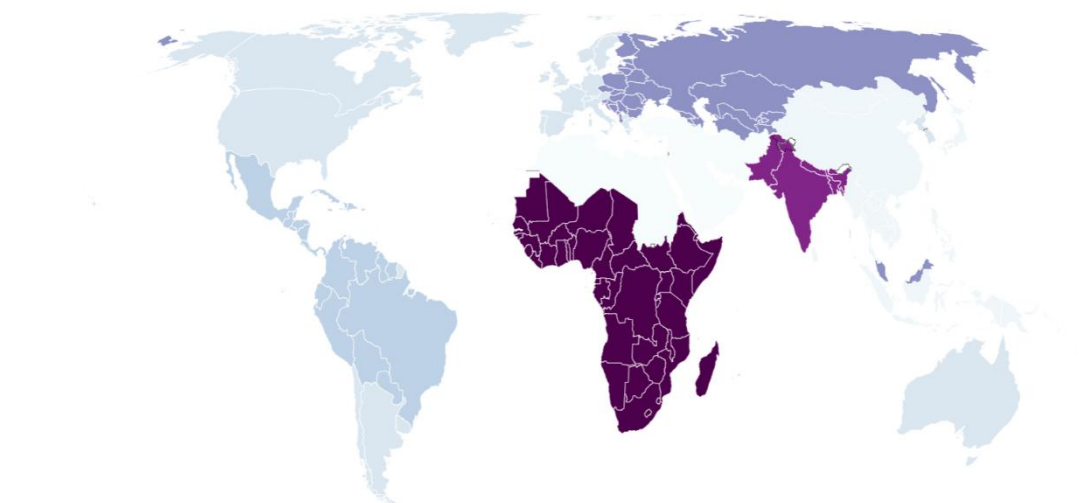
Source: Hendriksen, R.S., Munk, P., Njage, P. et al. Global monitoring of antimicrobial resistance based on metagenomics analyses of urban sewage. Nat Commun 10, 1124 (2019).

*measured in wastewater in fragments per kilo base per million fragments

Figure 1. Global predictions of antimicrobial resistance (AMR) across the globe.⁵

Deaths to AMR (rate per 100 000)

11.2 23.7



Source: Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis - Murray, Christopher J L et al. The Lancet, Volume 399, Issue 10325, 629 - 655

Figure 2. Death attributed to AMR resistance globally -Expressed in rates per 100 000.²

While the social and clinical importance of antibiotics is indisputable, their development faces a harsh economic reality. At the core of the issue lies a paradox: Antibiotics must be used judiciously to preserve their effectiveness and slow down the development of resistance, but this very restriction undermines their commercial viability. Unlike treatments for chronic or high-revenue conditions like cancer, antibiotics are typically prescribed for short courses and, in the case of new

ones, reserved as a last resort. This creates a market where demand is intentionally suppressed, an unthinkable scenario in pharmaceuticals, making antibiotics financially unattractive to large investors and drug developers.

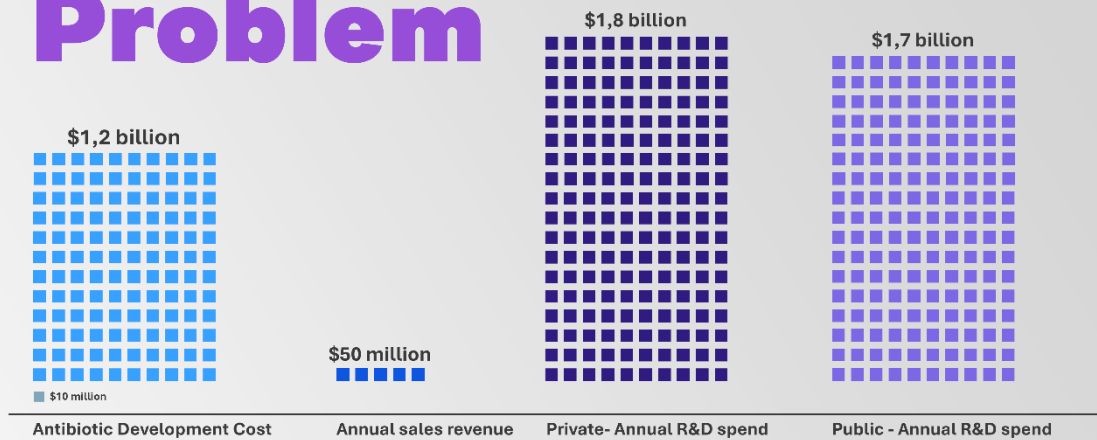
This structural problem has driven many major pharmaceutical companies away from antibacterial R&D. Since the 1990s, 18 of the largest pharmaceutical firms have exited the field entirely.⁶ Even among those that remained active longer, such as GSK, Novartis, Sanofi, Pfizer, and AstraZeneca, all eventually scaled down or discontinued their antibiotic programs between 2016 and 2019. What remains of antibiotic innovation today is largely carried by academia and small to medium-sized enterprises (SMEs). Unfortunately, these smaller players often lack the financial resilience to navigate the long, expensive path to market.⁷

Several high-profile collapses illustrate just how broken the current antibiotic market is. Tetrphase Pharmaceuticals, for example, invested more than a decade into developing eravacycline (Xerava®), which was approved by the FDA in 2018.⁸ Despite its medical value, poor sales led to massive financial losses, and by 2020 the company was sold after a dramatic fall from a peak valuation of \$1.8 billion to essentially nothing within 5 years.⁹ A similar fate was shared by Achaogen, the developer of plazomicin (Zemdri®), a drug designed to treat infections caused by carbapenem-resistant Enterobacteriaceae (CRE).¹⁰ Though approved by the FDA in 2018, the drug earned under \$1 million in sales, and the company filed for bankruptcy just a year later.

The list continues. In 2022, Spero Therapeutics, which had been developing a promising treatment for drug-resistant urinary tract infections (UTI) was forced to lay off 75% of its workforce after the FDA withheld approval of its Phase 3 candidate.¹¹ Spero's struggles highlight a larger trend: even companies that reach the final stages of development or gain approval can fail to survive the current landscape of antibiotic research and commercialization.

These outcomes underscore the severe market imbalance between cost and return. For context, the median cost to develop a cancer drug in 2017 was estimated at \$640 million, with typical post-approval revenues of \$1.7 billion.¹² In contrast, developing a new antibiotic can cost around \$1.2 billion.¹³ Nonetheless, antibiotics launched between 2010 and 2020 generated an average of just \$46–50 million in annual revenue, with total cumulative sales across all these drugs reaching only \$714.3 million, less than the yearly revenue of a single blockbuster drug.¹⁴

The Money Problem



\$3.4
Trillion

Is the predicted impact of
AMR infections on the
global economy by 2030

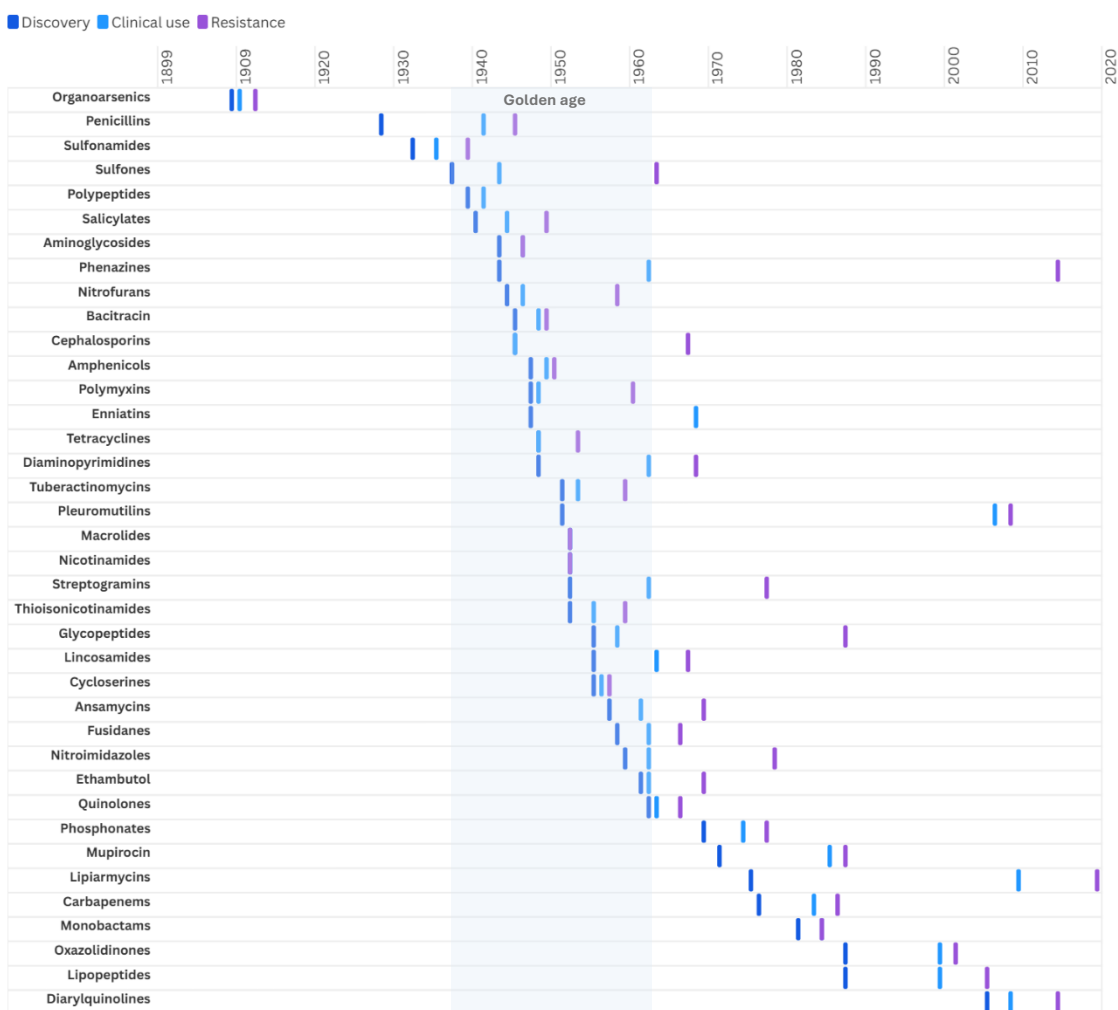
For comparison, Merck's cancer immunotherapy Keytruda (pembrolizumab) generated \$29.5 billion in 2024 alone, while Novo Nordisk's diabetes drug Ozempic (semaglutide) brought in \$17.5 billion that same year.¹⁵

Unsurprisingly, venture capital investment has followed this disparity, antimicrobial R&D attracted only \$1.6 billion over the decade leading to 2022, contrasting with the \$26.5 billion invested in oncology during the same period⁹

Adding to the devastating toll in death and disability, current underinvestment in antibiotic innovation is dwarfed by the projected economic impact of drug-resistant infections. The World Bank estimates that antimicrobial resistance could lead to an additional \$1 trillion in healthcare costs by 2050, and annual global GDP losses ranging from \$1 trillion to \$3.4 trillion as early as 2030.¹⁶

The stagnation in antibiotic innovation is not only a market failure, but also reflects the scientific challenges associated with antibiotic discovery (**Figure 3**). Since the so-called “golden age” of antibiotic discovery (1940s–1960s), which yielded most of the classes in use today, the pipeline for truly novel antibiotics has slowed dramatically¹⁷

Antibiotic discovery timeline



Source: Stennett, H.L.; Back, C.R.; Race, P.R. Derivation of a Precise and Consistent Timeline for Antibiotic Development. *Antibiotics* 2022, 11, 1237.

Figure 3. Timeline of antibiotic discovery, clinical use, and resistance.¹⁷

Remarkably, since the 1980s, only one entirely new class of antibiotics, the diarylquinolines, used to treat multidrug-resistant tuberculosis, has been introduced to the market.¹⁷

Despite advances in automation and the use of high-throughput screening (HTS) technologies, these approaches have largely failed to produce new, effective antibiotic scaffolds.¹⁸ The complex biology of bacterial pathogens, especially Gram-negative bacteria,¹³ and the physicochemical properties required¹⁹ for antibiotic activity limit the success of conventional HTS platforms. Natural products (NPs), historically a rich source of antibiotic compounds, offer promising bioactivity but come with their own limitations: they are often difficult to synthesize or modify, and the rediscovery of known molecules remains a major obstacle.^{20,21} As a result, innovation has been incremental rather than transformational, with many “new” antibiotics being variations of existing classes rather than genuine breakthroughs.

To grasp the dimension of the challenge we face, consider this: According to the European Space Agency,²² the Milky Way contains roughly 100 billion stars. Meanwhile, the chemical space of drug-like molecules is estimated at 10^{60} , an astronomically vast number.²³ Now, compare that with bacterial reproduction: a single bacterium, doubling every 20 minutes,²⁴ would reach that same number in just under 3 days. In contrast, it takes us 10 to 12 years to bring a single antibiotic to market, if we succeed at all.¹⁹ In other words, bacteria can explore evolutionary space faster than we can even begin to test the entirety of the available chemical space. Our problem is not the lack of possible antibiotics; it is the speed, scale, and structure of how we discover and develop them.

Aim and Scope

In this context, machine learning (ML) and drug repurposing have emerged as powerful strategies to accelerate antibiotic discovery. By using computational models to navigate vast chemical spaces more intelligently and by re-evaluating approved drugs for antibacterial potential, these approaches offer a promising path to outpace resistance and reinvigorate a stagnant pipeline.

Drug repurposing offers a promising strategy to address the drought of new antibiotics by bypassing much of the early discovery and preclinical development pipeline. Because pharmacokinetic, pharmacodynamic, and toxicity profiles are often already established, repurposed drugs can skip or shorten several initial phases, including Phase I clinical trials.^{19,25} From an economic perspective, this can translate into cost savings of approximately 15% for early-stage reductions,²⁵ and up to 85% when considering the entire development process.²⁶ Given that de novo antibiotic development can exceed \$800 million, this represents a substantial economic advantage. Moreover, repurposing reduces the risk of failure: approximately 45% of traditional drug candidates fail due to toxicity and safety issues.¹⁹ In contrast, repurposed drugs typically follow a faster and more efficient path to market, requiring an estimated 3 to 8 years, compared to 12 to 17 years for novel compounds.¹⁹

Regarding Deep learning approaches, they have significantly advanced our ability to explore vast chemical spaces.^{27,27,28} The integration of machine learning into drug discovery pipelines has enabled the rational design of bioactive compounds, some of which have demonstrated efficacy in animal models.^{29,30} These advances suggest that computational tools can go beyond theoretical predictions to deliver viable preclinical antibiotic candidates. This potential was exemplified by the discovery of Halicin,²⁰ a novel antibiotic identified using deep neural networks, which showed broad-spectrum activity against *Acinetobacter baumannii*, showcasing the ability of AI to uncover narrow-spectrum antibiotics with high specificity, including against multidrug-resistant pathogens.

Although the use of machine learning in drug discovery has grown substantially, its application remains uneven across therapeutic areas. Nunez,³¹ revealed that publications linking ML with cancer therapies or cardiovascular drugs vastly outnumber those focused on antibiotics. This discrepancy highlights a research gap, despite the urgent global need for novel antibiotics to combat antimicrobial resistance. Expanding the application of ML to antibiotic discovery thus represents both a challenge and an opportunity.

This project aims to leverage machine learning and cheminformatics techniques to identify promising antibiotic-like candidates within a curated set of FDA-approved non-antibiotic compounds. Although direct potency prediction (e.g., MIC regression) was initially considered, inconsistencies in bioassay data led us to adopt a more robust and interpretable strategy: structural similarity modeling. By combining circular fingerprints (ECFP4), physicochemical descriptors, and functional group annotations, we trained similarity-based models (e.g., k-nearest neighbors) to detect compounds with high chemical resemblance to known antibiotics. Dimensionality reduction using UMAP was applied to visualize and explore the chemical space, while clustering and probability scoring allowed prioritization of potential hits. This data-driven approach contributes to the broader effort of accelerating antibiotic discovery, offering a scalable and reproducible framework for candidate selection in the fight against AMR.

Database Search

To support this work, [ChEMBL](#), a comprehensive database of bioactive, drug-like small molecules was used. ChEMBL provides 2D molecular structures, calculated physicochemical properties (such as molecular weight and logP), and curated bioactivity data extracted from primary scientific literature. It includes structure-activity relationship (SAR) information, binding constants, and pharmacokinetic data, making it a critical resource for drug discovery and repositioning studies. In this work, ChEMBL served as the primary source for both the set of known antibiotics and the FDA-approved non-antibiotic compounds, enabling systematic comparison and feature extraction for machine learning modeling.

To start our analysis, the organism selected was *Escherichia coli* (*E. coli*). *E. coli* is the most extensively studied microbial organism in the world, making it an ideal model for antibiotic discovery. It is a Gram-negative, anaerobic bacillus belonging to the *Enterobacteriaceae* family. Clinically, *E. coli* is highly relevant due to its diverse pathogenic variants (pathotypes), including uropathogenic (*UPEC*), enterotoxigenic (*ETEC*), and enterohemorrhagic (*EHEC*) strains, each associated with distinct infection types, tissue targets, and mechanisms of pathogenesis.³² Notably, shiga toxin-producing *E. coli* (*STEC/EHEC*) alone comprises over 400 serotypes, illustrating the organism's genetic and phenotypic diversity.³³ This

diversity, combined with its global prevalence and well-characterized biology, makes *E. coli* an excellent representative for studying bacterial resistance and evaluating novel antimicrobial compounds.

Accordingly, ChEMBL was filtered for FDA-approved antibiotics (Max Phase = 4) with *E. coli* listed as a target organism. This search yielded a dataset (ab_raw) containing 25,092 entries across 48 variables.

Data Cleaning and Feature Engineering

Several columns were removed from the original dataset to improve consistency, reduce noise, and retain only relevant information for the project's objective. The criteria for removal were based on missing values, redundancy, inconsistency, or irrelevance to the structural analysis of compounds.

Data set – ab_raw

- Empty or sparsely populated columns such as pChEMBL Values, Data Validity Comment, and Comment, were discarded due to the lack of content.
- Ligand Efficiency metrics, including BEI, LE, LLE, and SEI, were not available in the dataset and thus excluded, although they are valuable indicators of compound efficiency in other contexts.
- Columns related to MIC value interpretation (e.g., Standard Relation, Standard Value, Standard Type, Standard Units, UO_Units) were removed due to inconsistent usage and redundancy. While MIC values were initially retained as potential predictive labels, their interpretation varied considerably across assays.
- BioAssay Ontology (BAO) fields such as BAO Format ID and BAO Label were excluded due to inconsistency between ontology classifications and actual assay descriptions (e.g., assays labeled as “metabolic” were in fact growth inhibition studies).
- Since only *E. coli* was selected as the target organism, related columns such as Target Name, Target Organism, Target Type, and Target ChEMBL ID were redundant and removed.
- Assay-specific metadata (e.g., Assay Tissue, Assay Cell Type, Assay Parameters, Assay Variant Accession, Assay Subcellular Fraction) were excluded due to high heterogeneity across entries, which would hinder model interpretability.

- Compound identifiers such as Molecule Name, Compound Key, and other naming variations were removed. Only “Molecule ChEMBL ID” (renamed to chembl_id - for reference) and SMILES (for RDKit processing) were retained.
- AlogP was removed and later recalculated as logP, using RDKit for consistency.
- #RO5 Violations was excluded due to evolving views on drug-likeness, especially in the context of non-traditional antibiotics and repurposing efforts.³⁴
- Additional assay-related metadata (Cell ChEMBL ID, Action Type, Standard Text Value, Value) and source publication details (Source ID, Source Description, Document Journal, Document Year) were also removed as they were not directly relevant to the structural analysis or predictive tasks.
- From here, the data set was renamed to ab_raw_clean. Finally, unlike previous data preprocessing steps performed in Excel, duplicate entries based on ChEMBL id and rows with missing ChEMBL id were removed using Python. After this processing step, the final dataset of FDA-approved antibiotics (**df_ab**) consisted of 756 molecules and 25 variables (excluding ECFPs), including the is_antibiotic binary label, which was added later to distinguish between known antibiotics and other compounds in subsequent analyses.

As mentioned previously, due to inconsistency in assay metadata and experimental conditions, MIC values for the same antibiotic varied significantly across the dataset. Factors such as bacterial strain, media composition, readout method, and assay format were often undefined or heterogeneous. This variability undermined the comparability of potency data and compromised the relationship between molecular structure and MIC values.

Consequently, predicting MIC directly was considered unfeasible. Instead, the task was redefined as a structural similarity problem, aiming to identify non-antibiotic approved drugs with molecular features most closely resembling known antibiotics. This pivot not only mitigates the influence of assay variability but also aligns more robustly with the principles of drug repurposing, where structural proximity to known actives can highlight previously overlooked antimicrobial potential.³⁵

After “cleaning” our dataset, the following variables were created to enhance the amount of structural information within the dataset. Firstly, the variable is_antibiotic was created. This variable explicitly labels compounds that are known FDA-approved antibiotics (is_antibiotic = 1), distinguishing them from other FDA-approved compounds with no known antimicrobial activity (is_antibiotic = 0).

Although the objective of this project was not to train a traditional classification model, the presence of a reference class enabled the application of similarity-based approaches.

Initially, using RDKit, several key physicochemical descriptors were computed to better characterize the chemical space of the molecules. These include logP, number of hydrogen bond donors (num_h_donors), number of hydrogen bond acceptors (num_h_acceptors), number of rotatable bonds (num_rotatable_bonds), topological polar surface area (TPSA), and number of rings (num_rings). These molecular descriptors are frequently used in cheminformatics and drug discovery to assess drug-likeness and predict molecular behavior in biological systems.³⁶

Next, binary variables were created to capture the presence or absence of key functional groups within each molecule. These variables were derived from the SMILES representations using SMARTS pattern matching, targeting substructures frequently associated with bioactivity.^{20,37} The groups considered included: *carboxylic acid, ether, ester, amine, amide, ketone, aldehyde, nitrile, aromatic ring, haloalkene, sulfonamide, acryl halide, nitro, alcohol, and alkene*. These structural indicators served as interpretable chemical descriptors, complementing the molecular descriptors derived from RDKit.

In addition to the functional group indicators, ECFPs were also computed for molecular characterization.²⁰ While traditional topological fingerprints were primarily developed for substructure and similarity searching, ECFPs were specifically designed to support structure–activity modeling, making them highly effective in cheminformatics applications such as virtual screening, molecular clustering, and predictive modeling.³⁸

In this project, ECFPs were used to represent the molecular structures of both known antibiotics and FDA-approved non-antibiotic compounds in a format that could be numerically analyzed. These fingerprints capture substructural information (Substructures refer to local chemical motifs or fragments within a molecule, such as functional groups, aromatic rings, or small ring systems) by encoding circular atom neighborhoods within a specified radius. The configuration used was ECFP4, which corresponds to a radius of 2, meaning each atom's environment was evaluated up to two bonds away. This allows a practical balance between capturing relevant chemical features and maintaining model efficiency. Additionally, each fingerprint was encoded into a binary vector of 1024 bits, where each bit indicates the presence or absence of a specific molecular substructure.

The ECFP4 fingerprints also played a central computing Tanimoto³⁹ similarity between molecules, enabling similarity-based ranking and identification of non-antibiotic molecules with antibiotic-like structures.

Data set – non_ab_approved_raw

For the non-antibiotic dataset, data were retrieved from ChEMBL by filtering for approved drugs (i.e., max_phase = 4), resulting in an initial dataset with 32 columns and 2,479 entries. To ensure consistency and relevance for chemical structure-based analysis, all columns were removed except for the ChEMBL ID and SMILES representation. A crucial curation step followed, in which molecules not typically considered drugs, such as salts, diagnostic aids, chelating agents, acidifiers, neurotransmitters (e.g., dopamine), isolated amino acids, organic bases, vitamins, UV screens, supplements, and electrolyte replenishers, were excluded. This decision aligns with the focus on bioactive, drug-like small molecules with potential pharmacological relevance. Additionally, any entries annotated as antibacterial or anti-infective agents were removed to avoid contamination of the negative class.

Following dataset curation, a similar preprocessing pipeline to the one used for the ab_raw_clean dataset, was applied to the non_ab_approved_raw data using Python. From here, the data set was renamed to non_ab_approved_clean.

Duplicate and null values in the chembl_id field were removed. A binary target variable, is_antibiotic = 0, was introduced to label all entries as non-antibiotics. Using RDKit, a set of physicochemical descriptors was calculated for each molecule, including logP, molecular weight, number of hydrogen bond donors (num_h_donors), number of hydrogen bond acceptors (num_h_acceptors), number of rotatable bonds (num_rotatable_bonds), topological polar surface area (TPSA), and number of rings (num_rings). Furthermore, binary variables indicating the presence of functional groups (e.g., carboxylic acid, ether, amine, etc.) were created using SMARTS patterns. Finally, ECFPs were computed using the same parameters applied to the antibiotic dataset (radius = 2, nBits = 1024), ensuring consistent structural representation across both datasets. After this processing step, the final dataset of FDA-approved non-antibiotics (**df_non_ab**) consisted of 1506 molecules and 25 variables (excluding ECFPs), including the is_antibiotic binary label.

Exploratory Data Analysis (EDA)

For the df_ab, based on the distribution plots (**Figure 4**) and correlation heatmap (**Figure 5**), several important observations can be drawn regarding the physicochemical properties and structural characteristics of known antibiotics:

First, the distribution of logP values (a measure of hydrophobicity) among antibiotics is centered around low (-2) to moderate values (4), indicating a preference for balanced hydrophilic/lipophilic profiles. Similarly, the molecular weight distribution concentrated below 600 Dalton (Da), consistent with typical

drug-like properties (e.g., Lipinski's Rule of Five).⁴⁰ These observations suggest that most antibiotics occupy a chemically constrained space, favoring, as expected, properties compatible with bioavailability and cellular penetration.

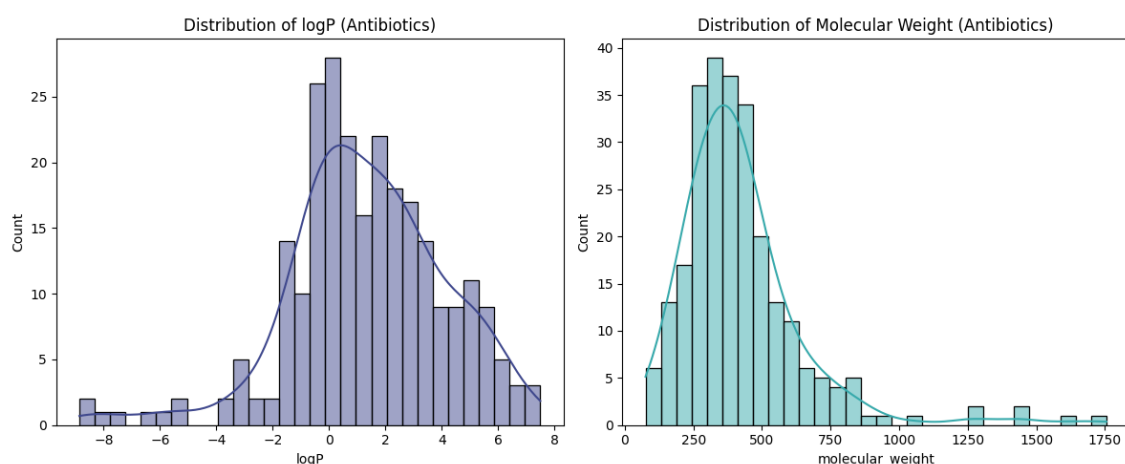


Figure 4. Distribution of $\log P$ and molecular weight among FDA-approved antibiotics.

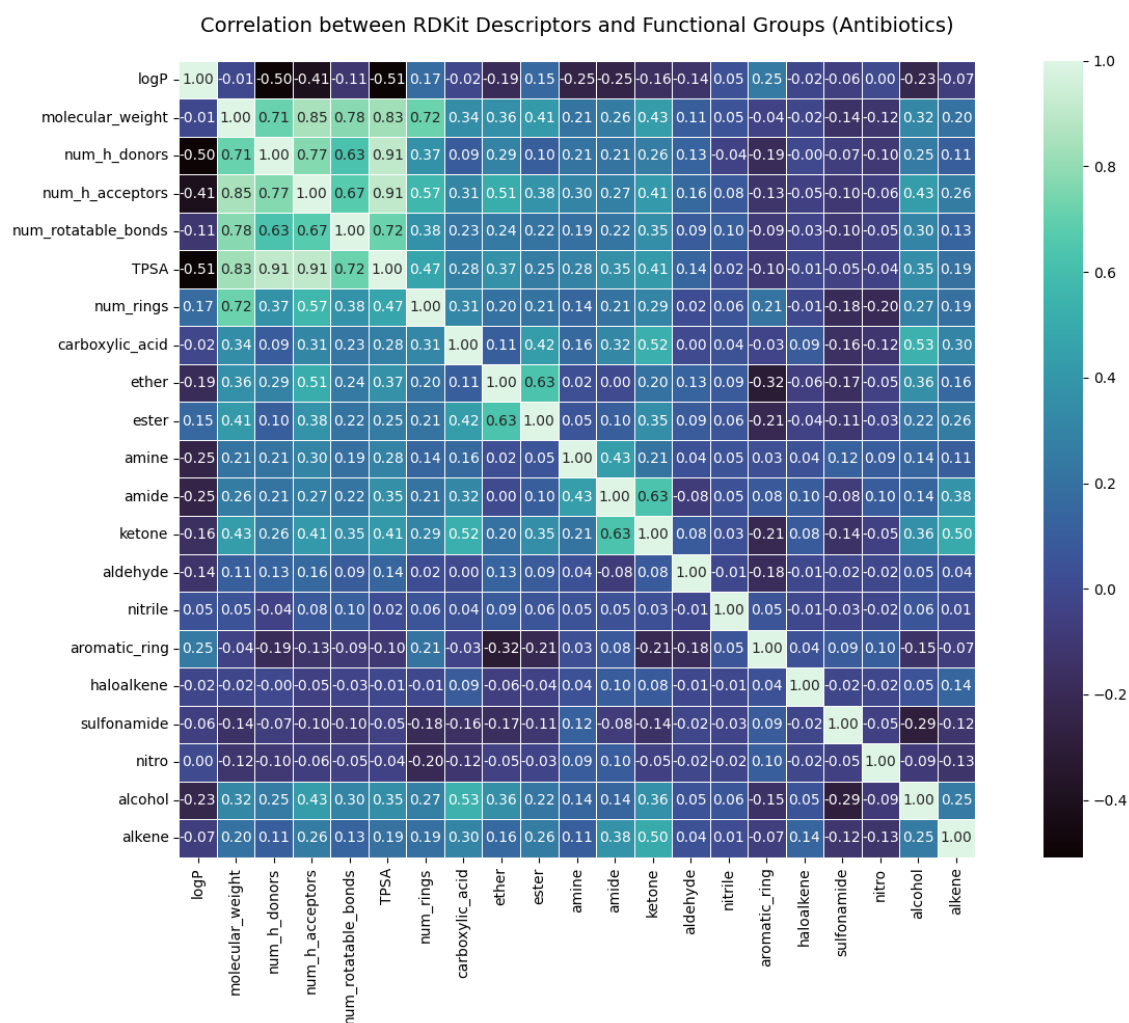


Figure 5. Correlation matrix between RDKit-calculated descriptors and functional group presence among antibiotics

The functional group analysis (**Figure 6**) reveals that the most frequent moieties in antibiotic molecules include amines, aromatic rings, alcohols, and ketones, all of which are commonly found in bioactive structures.³⁷ In contrast, groups like acryl halide, aldehyde, and nitrile are rarely or never present. Unsurprisingly, the correlation heatmap further confirms that descriptors such as molecular weight, TPSA, and hydrogen bonding capacity are moderately correlated with specific functional groups. For instance, alcohol and carboxylic acids are associated with higher TPSA and donor counts, while aromatic rings contribute to increased molecular weight and rigidity (ring count).

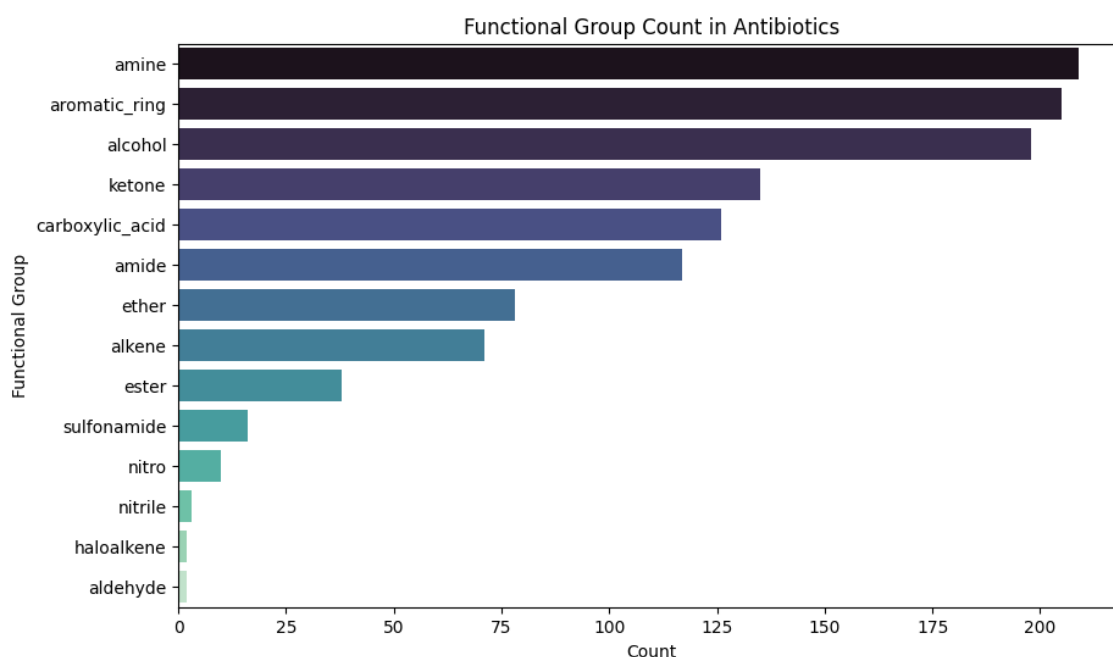


Figure 6. Functional group count on FDA-approved antibiotics

Regarding the observed patterns, for the non_ab_approved data set, the distribution of logP values is slightly shifted toward more lipophilic compounds compared to antibiotics, with most values centered in the range of 1 to 4. This reflects a general tendency for non-antibiotic drugs to favor membrane permeability and systemic distribution. The molecular weight distribution, although also right-skewed, appears more tightly clustered around lower values, suggesting that simpler scaffolds with fewer heavy atoms or less peptide-like structures than those typically found in antibiotics.

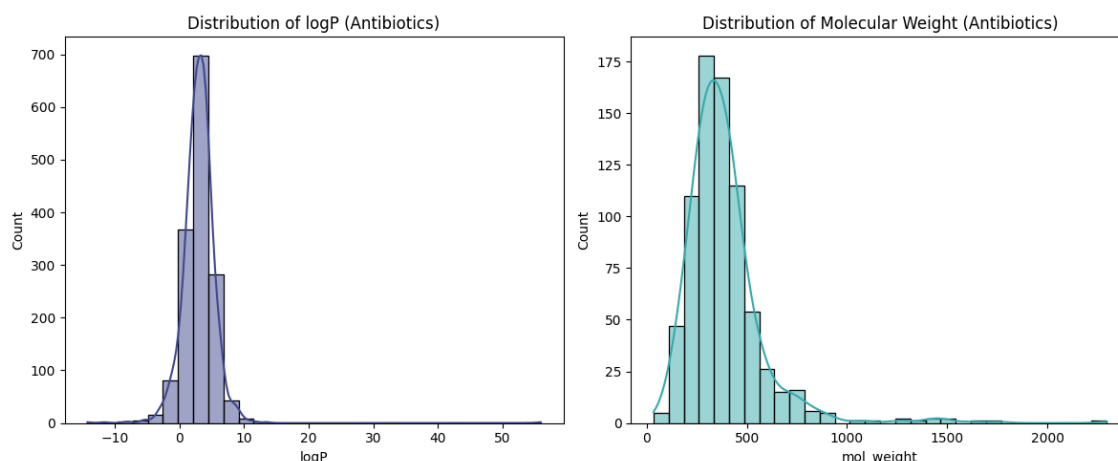


Figure 7. Distribution of logP and molecular weight among

The functional group frequency reveals a strong prevalence of aromatic rings, amines, alcohols, and ketones, which are also common in antibiotics, indicating shared drug-like features. However, groups like esters, ethers, and alkene moieties appear more frequently in non-antibiotics, whereas more polar or reactive groups (e.g., sulfonamides, nitro, or carboxylic acids) are less dominant. These distinctions may reflect different modes of action or metabolic requirements. Notably, as with antibiotics, rare functional groups like aldehydes and acryl halides remain mostly absent.

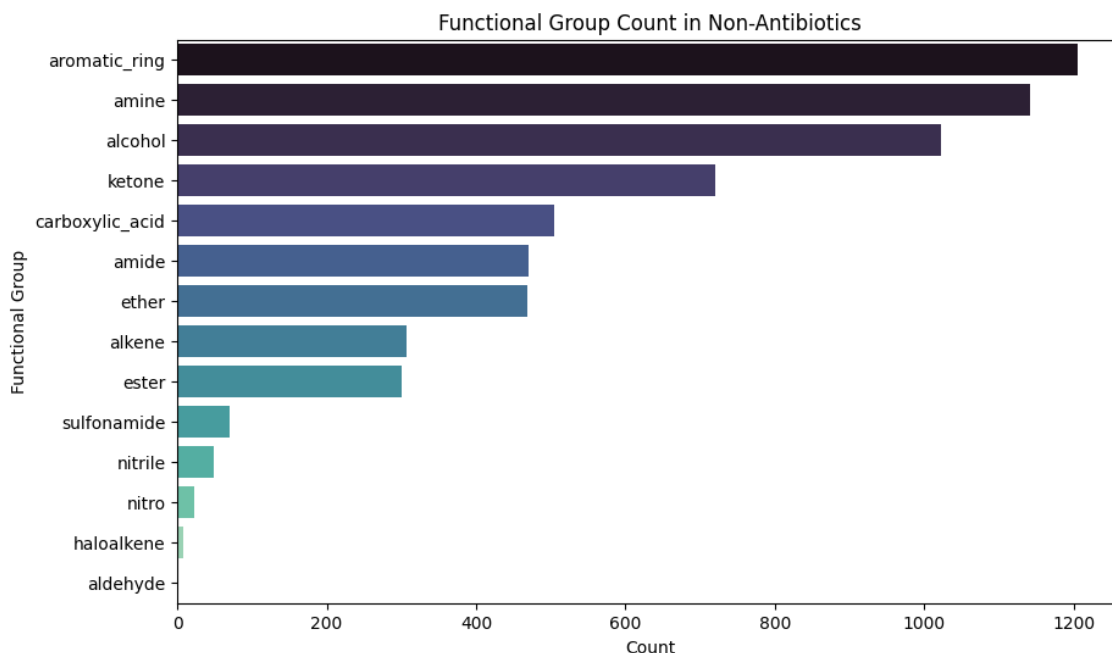


Figure 8. Functional group count on FDA-approved non-antibiotics

The correlation analysis between functional groups and RDKit descriptors shows weaker but still informative relationships. Molecular weight and TPSA (expectedly) continue to correlate positively with groups like carboxylic acids and alcohols, while aromatic content is linked to increased ring counts. However, compared to

antibiotics, these correlations tend to be less pronounced, possibly due to a narrower chemical space or reduced functional group diversity. Altogether, these insights underscore a distinguishable chemical fingerprint between antibiotic and non-antibiotic compounds.

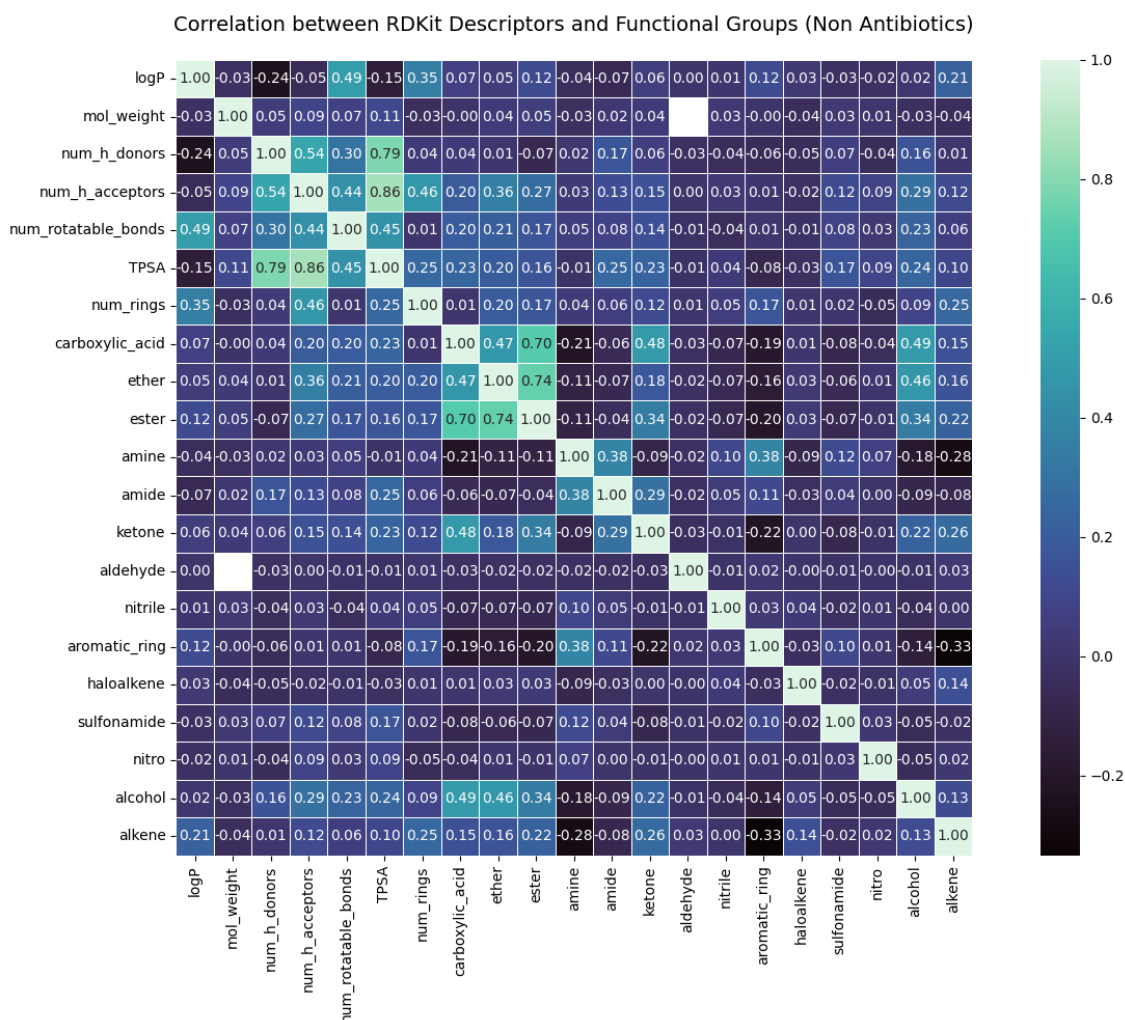


Figure 9. Correlation matrix between RDKit-calculated descriptors and functional group presence among non-antibiotics

Methodology

Initially, the two datasets, one consisting of known FDA-approved antibiotics and another comprising FDA-approved molecules not classified as antibiotics, were combined. Each molecule was characterized by a set of global physicochemical and structural descriptors, such as molecular weight, logP, hydrogen bond donors and acceptors, TPSA, and the presence of key functional groups (e.g., amide, ether, nitrile).

To visualize and explore the chemical space, Uniform Manifold Approximation and Projection (UMAP) was applied, a nonlinear dimensionality reduction technique that preserves local relationships.⁴¹ Features were standardized using StandardScaler, and UMAP was performed on the resulting matrix to obtain a 2D

Using the 2D UMAP embedding, a k-Nearest Neighbors (k-NN) classifier was trained to predict, based on physicochemical and structural descriptors, the likelihood that a given molecule belongs to the antibiotic class. The classifier was trained on both known antibiotics and non-antibiotics, and the predicted class probabilities were computed for each non-antibiotic compound. Molecules with an antibiotic-like probability above 0.80 were considered strong candidates for further investigation.

This approach leverages the topological structure captured by UMAP to identify regions in chemical space densely populated by antibiotics and identify close non-antibiotics as potential hits.

To further investigate the structural distribution of compounds, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) on the same UMAP embedding was applied. DBSCAN groups together dense regions and assigns a cluster label or marks a point as noise. This helped to reveal whether the high-probability candidates from k-NN fell within consistent local clusters or were scattered anomalies. It also informed the hybrid scoring scheme by providing a measure of structural density (i.e., whether a compound is part of a meaningful cluster).

To complement the previous descriptors, ECFP4 was extracted for all non-antibiotic compounds. These fingerprints represent the presence or absence of specific substructures within a molecule, capturing atom-level connectivity patterns up to a radius of 2 bonds.

We computed pairwise Tanimoto distances between ECFP4 fingerprints to quantify substructural similarity. This binary similarity metric measures the intersection over union of active bits in the fingerprint vectors. We used the resulting distance matrix to perform DBSCAN clustering, grouping molecules into structurally similar clusters based on shared chemical motifs.

By mapping the UMAP-derived k-NN probability with the substructure-informed cluster label, we obtained a richer representation of the candidate space, accounting for both global physicochemical similarity and detailed substructure features.

To prioritize candidates, we computed a hybrid score for each non-antibiotic compound, calculated as a weighted average of:

- The k-NN-based probability of being antibiotic-like (80% weight)
- A binary indicator of whether the compound was part of a dense ECFP4 cluster (20% weight)

This scoring method balances the probabilistic inference from UMAP with structural cohesion from fingerprint clustering. A higher weight for clustering was assessed

but led to the overrepresentation of alkaloid derivatives (e.g., *Vinca rosea* analogues), likely due their tendency to form large, dense clusters regardless of antibiotic-likeness. The 0.8 / 0.2 weighting provided a more interpretable and chemically relevant selection.

The final top-ranked candidates were visualized using RDKit, with molecular structures annotated by their probability or hybrid score. These candidates were then compared to known antibiotics in terms of logP and molecular weight distribution, and frequency of functional groups

Results

The dimensionality reduction performed via UMAP (**Figure 10**) revealed a distinguishable separation between known antibiotics and FDA-approved non-antibiotics in the 2D chemical space. Antibiotic molecules were predominantly located in a few well-defined regions, indicating a relatively constrained and clustered distribution in terms of their physicochemical and structural features. In contrast, the non-antibiotic molecules were more widely dispersed, with some lying close to or within antibiotic-enriched regions.

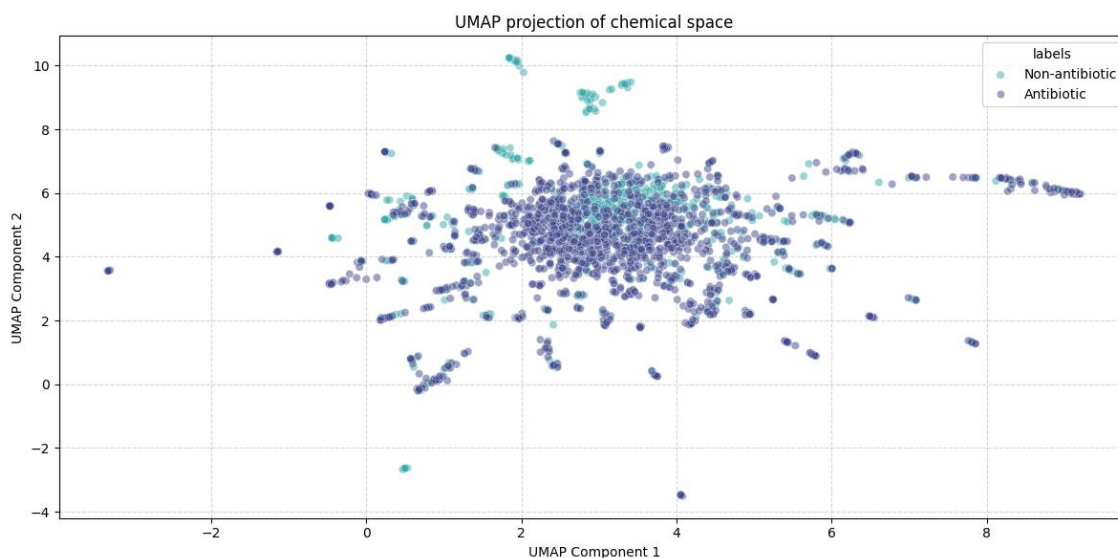


Figure 10. UMAP projection of the antibiotic and non-antibiotic chemical space based on global descriptors. Each point represents a molecule (either a known antibiotic or FDA-approved non-antibiotic), embedded into two dimensions using UMAP applied to standardized physicochemical and structural descriptors. Antibiotics (purple) tend to form dense clusters, while non-antibiotics (turquoise) are more dispersed, highlighting underlying separability in the reduced feature space.

Using the k-NN classifier on the UMAP-embedded space, four non-antibiotic compounds were identified as high-confidence antibiotic-like candidates (**Figure 11**), each exceeding the probability threshold of 0.80.

K-NN top candidates (prob. > 80%)

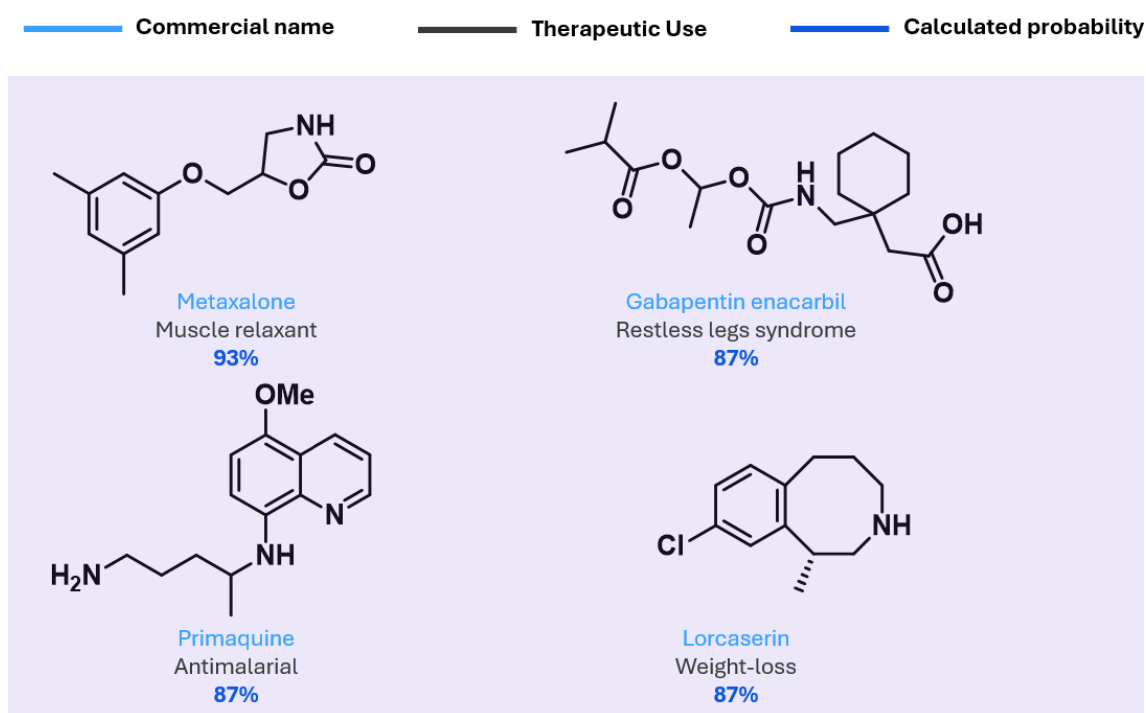


Figure 11. Top-ranked antibiotic-like candidates identified via *k*-NN classification in UMAP space. Four FDA-approved non-antibiotic compounds were predicted to have a high probability (> 80%) of being structurally similar to known antibiotics based on global physicochemical and functional group descriptors.

These compounds were visualized and annotated with their ChEMBL IDs and respective probability scores. While the small number of hits limits statistical comparisons, their proximity to dense antibiotic regions in UMAP space supports their potential relevance as structural analogues.

To assess whether these candidate molecules were part of structurally meaningful clusters, DBSCAN was applied to the UMAP coordinates. All four candidates were located within local clusters rather than as isolated outliers, suggesting that they share common structural patterns with neighboring molecules in the embedded space (**Figure 12**). This observation justified the incorporation of a clustering-based indicator into the hybrid scoring system.

A complementary DBSCAN clustering based on ECFP4 fingerprints, using Tanimoto distance as a substructure similarity measure, revealed that several non-antibiotic molecules formed dense clusters characterized by shared chemical scaffolds. Interestingly, when the hybrid score emphasized clustering too heavily (i.e., assigning more than 20% weight), the top-ranked candidates were dominated by alkaloid derivatives such as analogues of *Vinca rosea*. These molecules, while structurally consistent, may not necessarily share antibiotic-relevant mechanisms, highlighting the risk of overfitting to structural density alone. The final weighting scheme of 0.8 (UMAP *k*-NN probability) and 0.2 (cluster member) provided a balanced and interpretable ranking.

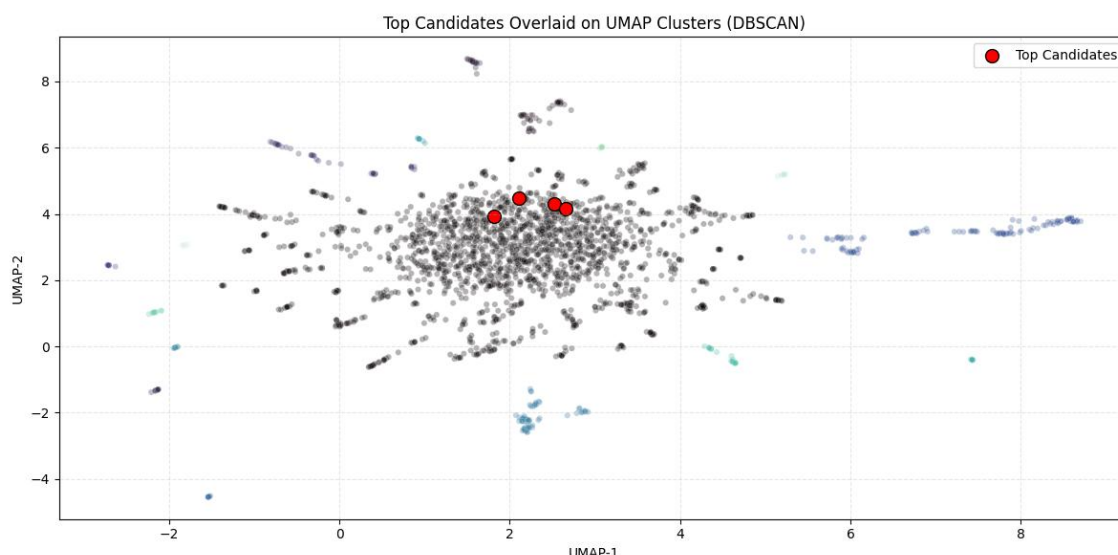


Figure 12. Top k -NN Candidates Localized within DBSCAN-Defined Structural Clusters

A comparative analysis of molecular features showed that the top candidates shared a similar profile with antibiotics in terms of logP and molecular weight (**Table 1**). However, due to the small number of candidates ($n=4$), rather than plotting histograms, we presented the raw values alongside the average of the antibiotic set. This provided a clearer and more direct comparison, confirming that all four candidates lie within the typical range of known antibiotics.

In terms of functional group composition, the candidates exhibited high similarity to antibiotics, particularly in the presence of amide, ether, and aromatic ring functionalities. This suggests that the model effectively captured substructural features often associated with antibiotic activity.

Table 1. logP and molecular weight values for the four top candidates identified by the k -NN classifier, compared to the average properties of the antibiotic dataset.

chembl_id	logP	molecular_weight (Da)
CHEMBL1079604	1.79	221.26
CHEMBL506	2.78	259.35
CHEMBL1628502	2.68	329.39
CHEMBL360328	2.59	195.69
Mean (Top Candidates)	2.46	251.42
Mean (Antibiotics)	1.73	386.60

Altogether, the combination of UMAP-guided k -NN prediction and ECFP4-based clustering enabled the identification of structurally plausible antibiotic-like

molecules. While experimental validation is needed, these hits demonstrate the potential of unsupervised and semi-supervised machine learning approaches to uncover drug repurposing opportunities from non-antibiotic libraries.

Future Perspectives

While this project demonstrates the potential of machine learning to identify antibiotic-like candidates through chemical similarity and substructure-based clustering, several critical limitations and opportunities for future development remain.

One important direction is the integration of genomic information. Current approaches rely solely on molecular descriptors and structural fingerprints, without considering the biological targets or mechanisms of action of antibiotics. Incorporating bacterial genomic data, such as conserved gene sequences or resistance pathways, could enable the development of target-specific antibiotics that act on essential and non-mutating regions of bacterial genomes. This would not only improve the precision of the predictions but also help avoid the rapid development of resistance.

In this context, a shift from broad-spectrum antibiotics to narrow-spectrum, targeted therapies may offer significant advantages. While broad-spectrum agents can be effective against a range of pathogens, they also disrupt the host microbiota and exert strong selective pressure, accelerating resistance. In contrast, narrow-spectrum drugs targeting specific bacterial species or strains could reduce collateral damage and extend the clinical lifespan of new therapies.

Another limitation of our approach lies in the exploration of chemical space. The methods applied here were based on repurposing known FDA-approved compounds, which limits the novelty of the discovered scaffolds. Although drug repurposing is efficient and cost-effective, it cannot fully address the urgent need for structurally novel antibiotics with new modes of action. To address this, future studies could explore de novo drug design, generative models, or virtual libraries that expand the accessible chemical space.

Even within the space of known molecules, further functional exploration is warranted. For instance, introducing or modifying functional groups ("tagging") on promising scaffolds could enhance activity, selectivity, or pharmacokinetics. Similarly, the exploration of bioisosteres (alternative atoms or groups with similar physicochemical properties) could yield derivatives with improved properties or even unveil new mechanisms of action while preserving activity.

An additional and very practical limitation encountered in this work relates to data availability and standardization. Despite growing efforts by the scientific

community to make chemical and bioactivity data publicly available through resources such as ChEMBL, [PubChem](#), or [ZINC](#), inconsistencies in data formatting, annotation, and metadata often pose challenges. In this project, considerable time was spent resolving mismatches in molecular identifiers (e.g., SMILES strings, ChEMBL IDs), aligning feature sets, and dealing with missing or heterogeneous descriptor data. These issues highlight the need for greater adherence to FAIR principles (Findable, Accessible, Interoperable, Reusable) in cheminformatics and bioinformatics.⁴²

In conclusion, future iterations of this work could benefit from a more integrated pipeline that combines chemical, biological, and genomic data, employs more advanced ML architectures, and prioritizes diversity-oriented exploration of chemical space, while also taking advantage of emerging tools for data harmonization and ontology-based annotation.

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